

Reva Hirave

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I study how sociotechnical systems shape human behavior and social outcomes, with a focus on online communities, social media, and digital well-being. I use computational social science, natural language processing, and causal inference to understand how platform design and online environments influence discourse, attitudes, and user behavior.

Keywords: Sociotechnical Systems, Computational Social Science, Human-Computer Interaction, Natural Language Processing, Causal Inference, Social Media, Online Communities, Digital Well-being

Education

Princeton University

Fall 2025 – Spring 2027

Fully-funded MSE, Computer Science

GPA: 3.75 / 4.00

Relevant Coursework: Applied Social Statistics, Machine Learning, AI for Humanists, Empirical Methods of Computing Research

George Mason University

Fall 2021 – Spring 2025

BSc, Applied Computer Science: Software Engineering

GPA: 3.81/4.0

Minors: Computational and Data Sciences, Web Design

Relevant Coursework: Advanced Natural Language Processing, Computational Social Science, Probability and Statistics for Engineers, Scientific Data and Databases, Database Systems, The Politics of Language

Thomas Jefferson High School for Science and Technology

Fall 2018 – Spring 2021

HS Diploma

- Mobile and Web Application Development Senior Research Lab

Teaching Experience

Graduate Teaching Assistantships

Computer Science: An Interdisciplinary Approach [COS 126]

Fall 2025 - Fall 2026

Undergraduate Teaching Assistantships

Industry-Sponsored Senior Design Project [CS 491]

Fall 2024

Data Structures and Algorithms [CS 310]

Spring 2023

Intro to Computational and Data Sciences Lab [CDS 102]

Fall 2022

Peer Mentorships

GMU LGBTQ+ Resources Center

Fall 2023

Professional Experience

Graduate Researcher

Humans and Machines Lab @ CITP @ Princeton University

Princeton, NJ

Aug 2025 - Present

Research Assistant (Data Engineering)

CLSP @ Johns Hopkins University

Baltimore, MD / Remote

Jun 2024 - Aug 2025

Undergraduate Research Assistant (NLP)

George Mason NLP

Fairfax, VA

Sep 2023 - May 2025

Research Intern (NLP)

KaDSci, LLC

Fairfax, VA

Sep 2024 - Dec 2024

IT Specialist Intern

RP Professional Services, LLC

Remote

Feb 2024 - May 2024

Software Architecture Intern

Zeta Associates, Inc

Fairfax, VA

Jun 2023 - Aug 2023

Design and Front-end Development Intern

Argot Nova, LLC

Remote

Dec 2022 - Aug 2023

Awards and Honors

Computer Science Departmental Honors	Spring 2025
Dean's List	Fall 2021 - Spring 2025
Undergraduate Research Scholars Program Grant	Spring 2023 - 2025
Break Through Tech AI Fellowship	Summer 2024
Outstanding BAM Student	Spring 2024
Goldwater Institutional Nominee	February 2024
Deltek Software Engineering Scholarship	January 2023

Technical Skills

Research Platforms & Full-Stack Development: Flutter, FastAPI, Google Cloud, Firebase, iOS development (Live Activities), Figma design and prototyping

Quantitative & Causal Methods: Mixed-effects regression, causal identification, natural experiment design, PyTorch, scikit-learn

Data Engineering: SQL, MongoDB, distributed/streaming pipelines, SLURM/HPC, Pandas, NumPy, Matplotlib

NLP Tools & Libraries: HuggingFace, Stanza, NLTK, WordNet

Programming Languages: Python, Java, C, R

Publications and Ongoing Projects

Publications

CrossWeave: Bridging Perspectives Across Online Communities with a Dual-Pane Design  *Salt Lake City, UT, USA
CSCW Companion '26*

Fang, Fei, **Hirave, Reva**, Jurayj, William, Li, Yuqi, Lu, Brian, Metin, Tarik, Miyake, Tsugunobu, Morhun, Kateryna, Permalla, Yash, Rustamov, Kenan, Shen, Allen, Shi, Haojun, Singh, Piccardi, Tiziano, Prabhav, Wang, Xiheng Tom, Xu, Kevin, Zeng, Qingcheng, Khashabi, Daniel, Perrin, Andrew J., Xiao, Ziang, Zhang, Jiayi, and Eisner, Jason M. (2026) (Authors listed alphabetically)

- CrossWeave is an AI-augmented social media interface that surfaces relevant posts from other conversation threads in a side pane, using LLM-identified highlights to help users discover cross-community perspectives and draft more constructive replies
- Built a multi-threaded data-mining pipeline to harvest and structure articles and comments from Reddit, Fox News, and The New York Times as the underlying dataset for the system's live demo

Who Let the Dogs Out? Auditing Dogwhistle Coverage in Hate Speech Benchmarks Across Target Groups  *EMNLP WOA 2026
Accepted*

Hirave, Reva and Oet, Ryan (2026)

- Audited dogwhistle coverage across two widely-used hate speech benchmarks (HateXplain, Measuring Hate Speech), mapping an existing 340-entry dogwhistle glossary onto a four-level coding sophistication taxonomy across 59.6k posts
- Identified annotation failure (mislabeling dogwhistle content as non-hateful) and collection failure (entire dogwhistle categories missing from the data) as distinct, unevenly distributed failure modes — e.g., race and LGBTQ+ subgroups showing the largest gaps — and released the mapped glossary and pipeline publicly

Ongoing Projects

Archively: Spillover Effects of TikTok Screen-Time Limits

Marianne Aubin Le Quéré, **Reva Hirave**, Tiffany Liu, Manoel Horta Ribeiro, Andy Guess

- Architected a full-stack research platform (Flutter iOS app, FastAPI backend on GCP/Firebase) to recruit and pseudonymously track participants uploading TikTok data exports and Screen Time records
- Investigating whether platform-imposed screen-time limits produce spillover effects onto other apps and overall digital wellbeing

Unscrambled: An EMA Platform for Studying Time Perception in Short-Form Video

Reva Hirave, Manoel Horta Ribeiro

- Designed a single-mechanic ecological momentary assessment platform collapsing measurement and intervention into a guess-then-feedback task on cumulative session time, delivered via iOS Live Activities
- Investigating how users' perceived elapsed time diverges from actual scrolling duration on short-form video platforms

Counterfactual Comments: A Cross-Platform Causal Design for Comment Toxicity

*Reva Hirave**, Mehedi Zaman*, Francesco Salvi*, Jason Greenfield*, Manoel Horta Ribeiro

**These authors contributed equally.*

- Investigating the causal effect of early comment-thread toxicity on subsequent toxicity, using a novel identification strategy to isolate toxicity spillover from content-level confounding
- Building a memory-safe streaming pipeline to extract and match comment threads across multi-terabyte Reddit data dumps on Princeton's HPC cluster

Pastcasting: Forecasting Historical Events with a Vintage LLM

Reva Hirave, Liv d'Aliberti, Manoel Horta Ribeiro

- Fine-tuning and instruction-tuning a 13B model with a 1931 training cutoff to test whether LLM forecasting accuracy degrades with temporal distance from training data
- Building a prediction-market-style benchmark generation pipeline spanning 1931-1977 historical Newswire text